Impact of Hybrid Structured Random Projections on Graph Embeddings

H.A.D.T.N. Haththella
Dept. of Computer Engineering
University of Peradeniya
Sri Lanka
e20133@eng.pdn.ac.lk

L.R.S.R. Premasiri
Dept. of Computer Engineering
University of Peradeniya
Sri Lanka
e20305@eng.pdn.ac.lk

A.N.T. Somathilaka
Dept. of Computer Engineering
University of Peradeniya
Sri Lanka
e20381@eng.pdn.ac.lk

Sampath Deegalla
Dept. of Computer Engineering
University of Peradeniya
Sri Lanka
sampath@eng.pdn.ac.lk

***Abstract*—Graph representation learning has become an important paradigm for analyzing networked data across domains such as social networks, citation graphs, and biological systems. Many scalable graph embedding approaches rely on dense Gaussian random projections, which can impose significant computational and memory overhead when applied to large graphs. This work investigates whether hybrid structured random projections can improve the trade-off between efficiency and predictive performance within the FastRP graph embedding framework.**

Three projection strategies are designed and compared: (i) a standard Gaussian projection as the baseline, (ii) a striped structured sparse projection intended to reduce density while preserving embedding quality, and (iii) a hybrid projection that concatenates a structural embedding component with a node feature component. Experiments are conducted on four benchmark datasets—BlogCatalog, Flickr, Cora, and ogbn-arxiv—covering node classification tasks, and on Cora, Citeseer, and PubMed for link prediction. Hyperparameters are optimized using Optuna with 50 trials per configuration, and downstream performance is evaluated using Macro-F1 score, classification accuracy, area under the ROC curve (AUC), and mean reciprocal rank (MRR@10).

On the Cora citation network, the hybrid projection achieves a Macro-F1 of 0.7541, outperforming the Gaussian baseline (0.7232) and the striped projection (0.7356). On the large-scale ogbn-arxiv benchmark, the hybrid approach yields a test accuracy of 0.6891 versus 0.5211 for the Gaussian baseline—a gain of 16.8 percentage points—demonstrating that integrating node feature information into the projection substantially improves embedding quality on feature-rich graphs. The striped structured projection is competitive on citation graphs but shows sensitivity to dataset characteristics on Flickr, where it trails the Gaussian baseline by a small margin. Overall, the results indicate that carefully designed hybrid structured projections are a promising direction for scalable graph representation learning.

***Index Terms*—Graph Embedding, FastRP, Random Projection, Hybrid Projection, Node Classification, Link Prediction, Graph Neural Networks, Optuna**

I. Introduction

The growing prevalence of graph-structured data in social networks, knowledge graphs, and citation databases has driven significant interest in graph representation learning. The central goal is to map each node to a low-dimensional vector that captures structural and semantic properties of the graph, enabling effective downstream tasks such as node classification and link prediction [1], [2].

Among the available approaches, Graph Neural Networks (GNNs) have become a dominant paradigm for learning node representations through iterative message passing over local neighborhoods [1]. However, end-to-end trained GNNs face scalability challenges on large real-world graphs: full neighborhood aggregation incurs memory and compute costs that scale with graph size, and gradient-based training across many epochs amplifies these costs [3].

Projection-based graph embedding methods offer a compelling alternative. By initializing node representations with random projections and propagating them across the graph, methods such as FastRP [4], RandNE [5], and SketchNE [6] achieve remarkable scalability without incurring the training overhead of deep architectures. The theoretical foundation for these approaches is the Johnson-Lindenstrauss (JL) lemma [7], which guarantees that pairwise distances can be approximately preserved after projection into a lower-dimensional space with high probability.

Despite their efficiency, conventional dense Gaussian random projections still impose non-trivial computational overhead, and recent analyses have shown that random projections can degrade node similarity measures under certain graph structures [8]. Sparse and structured projection matrices, as established by Achlioptas [9], can reduce this cost while maintaining JL guarantees.

Building on these observations, this work addresses the following research question: *Can hybrid structured random projections—combining sparse structural embeddings with node feature information—outperform a dense Gaussian baseline in terms of downstream node classification and link prediction accuracy, while remaining computationally tractable?*

The main contributions of this paper are:

- A clean, modular implementation of three projection strategies (Gaussian, Striped, Hybrid) within a unified FastRP-style pipeline.
- A systematic evaluation across four node-classification datasets and three link-prediction datasets, with Optuna-based hyperparameter tuning for fair comparison.
- Empirical evidence that the hybrid structure-feature projection substantially and consistently outperforms the Gaussian baseline on feature-rich graphs.
- An analysis of the conditions under which structured sparse projections succeed and where they remain sensitive.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the methodology. Section IV covers the experimental setup. Section V presents and discusses the results. Section VI concludes.

II. Related Work

A. Scalable Graph Embedding

Projection-free embedding methods that rely on random walks—including DeepWalk [10] and Node2vec [11]—apply Skip-Gram language modelling to graph traversals to learn latent node representations. DeepWalk achieves up to 10% higher F1 than spectral baselines on sparse-label benchmarks, but the walk generation step is expensive and difficult to parallelise at scale. Node2vec extends this with biased walks controlled by return and in-out parameters, yielding greater flexibility in capturing breadth-first versus depth-first neighbourhood structure [11].

To avoid walk generation, matrix-factorisation approaches such as LINE [12] optimise explicit proximity objectives for first- and second-order proximity, while NetSMF [13] approximates the implicit matrix objective of DeepWalk through sparse matrix factorisation, demonstrating that sparse operations are essential for billion-scale graphs. HOPE [14] addresses directed graphs by preserving asymmetric transitivity via high-order proximity measures such as the Katz index and rooted PageRank, showing that orientation-aware proximity cannot be recovered by symmetric-proximity methods alone.

Projection-based approaches eliminate both walk generation and matrix decomposition. RandNE [5] uses iterative random projections to preserve high-order proximity without expensive factorisations, scaling to billion-edge graphs. SketchNE [6] combines sketching with randomised SVD, embedding billion-node networks in under one hour and outperforming walk-based methods in both speed and quality. FastRP [4] reduces density further with very sparse projections and iterative propagation, achieving Macro-F1 scores competitive with walk-based methods while reducing runtime by orders of magnitude, confirming that projection-based pipelines can match walk-based quality at a fraction of the cost.

B. Structured, Sparse, and Hybrid Random Projections

The theoretical foundation for sparse projection matrices was established by Achlioptas [9], who proved that entries drawn from $\{-1, 0, +1\}$ with up to two-thirds zero density preserve pairwise distances under the Johnson-Lindenstrauss lemma [7]. This result implies that sparsity does not fundamentally degrade embedding quality and motivates the construction of structured sparse matrices that are both fast to apply and memory-efficient.

Yahaya et al. [15] formalised Hybrid Random Projections (HRP) as a mixture $\mathbf{H} = \alpha\mathbf{R} + \beta\mathbf{S}$, where $\mathbf{R}$ is a classical Gaussian matrix, $\mathbf{S}$ encodes structural constraints, and the scalars α, β govern the randomness–structure balance. Across several high-dimensional benchmark datasets, this formulation consistently reduces geometric distortion relative to purely Gaussian projections. Hierarchical feature selection combined with random projection [16] further refines projection pipelines by identifying salient dimensions before projection, reducing noise and memory overhead without expensive matrix inversion.

In the GNN context, RpHGNN [17] compresses heterogeneous graph tensors via random projection for efficient mini-batch training, reporting speedups exceeding 230% over conventional GNN training. RAP-GNN [3] replaces all learnable message-passing weights with diagonal random matrices sampled on-the-fly, cutting training time by a factor of six and memory by a factor of three while achieving accuracy comparable to fully trained GNNs—challenging the assumption that explicit parameter learning is always required. Random projection forest initialisation [18] improves GCN embedding quality by weighting edges according to co-occurrence frequency across multiple projection trees, producing higher-quality neighbourhood graphs than standard k-NN constructions.

Despite these advances, Tadíć et al. [8] document pathological cases in which random projections degrade node similarity and ranking quality, particularly for nodes at the extremes of the degree distribution, underscoring that projection design must be sensitive to graph topology.

C. Research Gap

Existing work either targets pure Gaussian projections (FastRP, RandNE, SketchNE) or integrates random projections into GNN training loops (RpHGNN, RAP-GNN). No prior study has conducted a controlled, multi-dataset comparison of (a) a standard Gaussian baseline, (b) a structured sparse projection inspired by Achlioptas [9], and (c) a hybrid projection that combines structural embedding with node feature information, all within the same FastRP-style pipeline and with identical hyperparameter tuning infrastructure. The present work fills this gap and provides the first systematic evidence of the conditions under which each projection type is beneficial.

III. Methodology

A. Problem Formulation

Let $G = (V, E)$ be an undirected graph with $N = |V|$ nodes and adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$. The row-normalized transition matrix is $\mathbf{T} = \mathbf{D}^{-1}\mathbf{A}$, where $\mathbf{D}$ is the diagonal degree matrix. Some datasets also provide a node feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$. The goal is to learn an embedding

matrix $\mathbf{Z} \in \mathbb{R}^{N \times d}$ ($d \ll N$) suitable for downstream node classification or link prediction.

B. Base Method: FastRP

Our pipeline builds directly on FastRP [4], which establishes the two-component design paradigm for scalable graph embedding: (1) constructing a node similarity matrix that preserves high-order graph proximity, and (2) applying an optimization-free dimension reduction method to that matrix.

a) Similarity matrix.: FastRP [4] observes that the k-step transition matrix $\mathbf{T}^k$ converges entrywise towards $d_j/2m$ as k grows—a consequence of the Perron–Frobenius theorem applied to the symmetrized transition matrix [4]. Because many real-world graphs are scale-free, this produces a heavy-tailed entry distribution that harms dimension reduction [19]. To correct for this, FastRP [4] applies a degree-based normalization derived from a Tukey power transform [20]: each column j of $\mathbf{T}^k$ is scaled by $(d_j/2m)^\beta$ for a tunable exponent $\beta \in [-1, 0]$, yielding the corrected matrix

$$\widetilde{\mathbf{T}}^k_{ij} = \mathbf{T}^k_{ij} \left(\frac{d_j}{2m} \right)^\beta . \tag{1}$$

This normalization downweights the influence of high-degree hub nodes and can be expressed as right-multiplication by a diagonal matrix $\mathbf{L}$, enabling fully iterative computation [4].

b) Very sparse random projection.: For dimension reduction, FastRP [4] draws the projection matrix $\mathbf{R} \in \mathbb{R}^{N \times d}$ from the very sparse distribution of Li et al. [19], where each entry is independently drawn from

$$R_{ij} = \begin{cases} +\sqrt{s}, & \text{with probability } \frac{1}{2s}, \\ 0, & \text{with probability } 1 - \frac{1}{s}, \\ -\sqrt{s}, & \text{with probability } \frac{1}{2s}, \end{cases} \tag{2}$$

with $s = \sqrt{m}$ recommended, achieving $\sqrt{m}$-fold speedup over dense Gaussian projection while preserving Johnson–Lindenstrauss guarantees [19], [21].

c) Iterative computation and weighted combination.: A key algorithmic contribution of FastRP [4] is the exploitation of matrix associativity to avoid forming $\mathbf{T}^k$ explicitly. Starting from $\mathbf{N}_1 = \mathbf{TLR}$, each subsequent order is obtained as $\mathbf{N}_i = \mathbf{TN}_{i-1}$ at cost $O(md)$ per step. The final embedding is a weighted sum over propagation orders:

$$\mathbf{N} = \sum_{i=1}^{k} \alpha_i \mathbf{N}_i, \tag{3}$$

where $\alpha_1, \ldots, \alpha_k$ are tunable scalar weights. This iterative scheme gives FastRP [4] overall time complexity $O((N + m)kd)$, linear in the graph size.

C. Our Implementation and Adaptations

We implement the above FastRP pipeline with the following adaptation to the normalization step. Rather than applying the exact $(d_j/2m)^\beta$ column scaling from Equation (1), we scale each row of the initial representation by d_i^α (fixed at $\alpha = -0.6$ in all experiments). This row-wise formulation is equivalent in intent—penalising the outward influence of high-degree nodes—but is applied before propagation rather than as a column scaling of $\mathbf{T}^k$. The exponent α plays the same role as FastRP's β and is kept fixed across experiments rather than tuned, in order to isolate the effect of the projection strategy.

Additionally, we apply ℓ_2-normalization to each intermediate representation $\mathbf{N}^{(t)}$ before accumulation, and apply a final per-dimension z-score standardization to the aggregated embedding $\mathbf{Z}$. Neither of these post-processing steps appears in the original FastRP paper [4]; they are introduced to improve numerical stability and conditioning of the downstream logistic regression classifier.

Formally, the pipeline as implemented proceeds as follows:

a) Step 1: Initial projection.: Construct the initial representation $\mathbf{N}^{(0)} \in \mathbb{R}^{N \times d}$ according to the chosen projection strategy (Section III-D).

b) Step 2: Degree-based row scaling (adapted from FastRP [4]).: Scale each row i of $\mathbf{N}^{(0)}$ by d_i^α, where $\alpha = -0.6$. This reduces hub-node dominance, analogously to FastRP's column normalization (1).

c) Step 3: Iterative neighborhood propagation (inherited from FastRP [4]).: For $t = 1, \ldots, k$:

$$\mathbf{N}^{(t)} = \mathbf{T}\,\mathbf{N}^{(t-1)}. \tag{4}$$

The embedding is accumulated as a weighted sum of ℓ_2-normalized intermediate representations:

$$\mathbf{Z}_{\text{raw}} = \sum_{t=1}^{k} w_t \cdot \frac{\mathbf{N}^{(t)}}{\|\mathbf{N}^{(t)}\|_2}. \tag{5}$$

d) Step 4: Final standardization (our addition).: The aggregated embedding is z-score standardized per dimension to produce the final output $\mathbf{Z}$.

D. Novel Projection Strategies

The central contribution of this work is the replacement of FastRP's very sparse projection [4], [19] with three alternative strategies for constructing $\mathbf{N}^{(0)}$, enabling a controlled study of how projection design affects downstream performance.

1) Gaussian Baseline: The projection matrix entries are drawn i.i.d. from a standard normal distribution:

$$R_{ij} \sim \mathcal{N}(0, 1). \tag{6}$$

Dense Gaussian projection is the classical reference for JL-type embeddings [21] and serves as the baseline against which sparser and hybrid strategies are compared. It is denser than FastRP's very sparse projection [4], [19] and incurs higher memory and compute cost.

2) Striped Structured Sparse Projection (Our Contribution): We propose a structured sparse projection in which the d output dimensions are partitioned into non-overlapping groups of size g (a tunable hyperparameter). For each node i and each group, exactly one column within the group receives a non-zero entry drawn uniformly from $\{-1, +1\}$, scaled by $\sqrt{g}$ to preserve the variance of the Gaussian baseline:

$$R_{ij} \in \{-\sqrt{g},\ 0,\ +\sqrt{g}\}, \quad \text{density} = \frac{1}{g}. \tag{7}$$

The group constraint enforces a regular non-zero pattern ("stripes") across the projection matrix, in contrast to the independent Bernoulli draws of Achlioptas [9] or Li et al. [19]. Larger g increases sparsity. JL guarantees are preserved at any density above $\Omega(\log N/d)$ [7], [9]. While the very sparse projection of FastRP achieves $s = \sqrt{m}$-fold speedup through independence across entries, our striped design introduces a structural regularity in the non-zero pattern that is motivated by the block-partition parallelization strategy described in [4].

3) Hybrid Structure-Feature Projection (Our Contribution): When node features $\mathbf{X}$ are available, we partition the d embedding dimensions into a structural component ($d_{\text{struct}} = \lfloor 3d/4 \rfloor$) filled by the striped projection, and a feature component ($d_{\text{feat}} = d - d_{\text{struct}}$) derived from the node features through a random Gaussian compression:

$$\mathbf{R}_{\text{feat}} = \mathbf{X}\,\mathbf{P}, \quad \mathbf{P} \in \mathbb{R}^{F \times d_{\text{feat}}}, \quad P_{ij} \sim \mathcal{N}\big(0,\, \tfrac{1}{F}\big)\,. \tag{8}$$

The two components are concatenated to form the initial representation:

$$\mathbf{N}^{(0)} = \big[\mathbf{R}_{\text{struct}} \parallel \mathbf{R}_{\text{feat}}\big]. \tag{9}$$

This design extends the FastRP pipeline [4]—which is purely topology-driven—by fusing node attribute information into the initial projection. The subsequent propagation steps then propagate both the structural and semantic signals jointly across the graph. The 3:1 structure-to-feature dimension ratio was chosen empirically; ablating this ratio is left as future work.

E. Downstream Tasks

a) Node Classification.: Embeddings are passed to a one-vs-rest logistic regression classifier (`liblinear` solver) trained on 80% of nodes and evaluated on the remaining 20%, measuring Macro-F1 score. For ogbn-arxiv, standard accuracy on the OGB split is also reported.

b) Link Prediction.: Edges are split into training (85%), validation (5%), and test (10%) sets. For each positive test edge, 500 negative edges are sampled. Link scores are computed as the dot product of node embeddings, and performance is measured by AUC and MRR@10 averaged over five independent random seeds. The model is tuned on the validation set and then evaluated on the test set.

IV. Experimental Setup

A. Datasets

Table I summarizes the benchmark datasets used.

TABLE I
BENCHMARK DATASETS

Dataset	Nodes	Edges	Features	Task
BlogCatalog	10,312	333,983	None	NC
Flickr	89,250	899,756	None	NC
Cora	2,708	5,429	1,433	NC / LP
ogbn-arxiv	169,343	1,166,243	128	NC
Citeseer	3,327	4,732	3,703	LP
PubMed	19,717	44,338	500	LP

NC = Node Classification; LP = Link Prediction.

BlogCatalog and Flickr are social/content-sharing networks loaded from standard `.mat` files. Cora, Citeseer, and PubMed are citation network benchmarks downloaded via PyTorch Geometric. ogbn-arxiv is loaded from the Open Graph Benchmark [22].

B. Implementation

The framework is implemented in Python using PyTorch for tensor operations and scipy for sparse matrix handling. The `FastRP` class supports all three projection types via a single `projection_type` argument. All experiments are executed with PyTorch `no_grad` (no backpropagation), so the approach is strictly unsupervised at the embedding stage.

C. Hyperparameter Tuning

Hyperparameters are tuned using Optuna [23] (TPE sampler) with 50 trials per experiment, optimizing Macro-F1 (node classification) or MRR@10 (link prediction) on a held-out validation split. The tuning space is:

- **Embedding dimension** $d \in \{64, 128, 256, 512\}$
- **Window size** $k \in [1, 5]$
- ℓ_2 **normalization** $\in \{\text{True}, \text{False}\}$
- **Group size** $g \in [2, 10]$ (striped and hybrid only)

The following parameters are fixed across all runs: $\alpha = -0.6$, propagation weights $\mathbf{w} = [1.0, 1.0, 7.81, 45.28]$ (derived from the BlogCatalog baseline), and `input_matrix = 'trans'` (row-normalized transition matrix). Final test-set results for link prediction are averaged over five random seeds (42, 100, 200, 300, 400) using the best hyperparameters from the validation phase.

V. Results and Discussion

A. Node Classification

1) Cora: Table II reports Macro-F1 scores on Cora. Both structured projection variants outperform the Gaussian baseline. The hybrid model achieves the highest score, demonstrating that incorporating node feature information (1,433-dimensional bag-of-words vectors) into the initial projection leads to measurably richer embeddings on this citation network.

TABLE II
NODE CLASSIFICATION — CORA (MACRO-F1)

Method	Macro-F1
Gaussian Baseline	0.7232
Striped Structured	0.7356
Hybrid	**0.7541**

2) ogbn-arxiv: Table III reports train, validation, and test accuracy on ogbn-arxiv. The Gaussian and striped projections achieve comparable scores ($\sim$0.52 test accuracy), performing poorly on this large feature-rich graph. The hybrid projection achieves a test accuracy of 0.6891—a gain of 16.8 percentage points over each of the structure-only baselines— highlighting

that the 128-dimensional paper embedding features in ogbn-arxiv carry strong discriminative signal that the structure-only methods cannot exploit.

TABLE III
NODE CLASSIFICATION — OGBN-ARXIV (ACCURACY)

Method	Train	Val	Test
Gaussian Baseline	0.5149	0.5080	0.5211
Striped HRP	0.5115	0.5063	0.5184
Hybrid HRP	**0.7468**	**0.6970**	**0.6891**

3) Flickr: Table IV reports Macro-F1 on Flickr, which has no node features, so only Gaussian and striped projections are directly comparable. The striped projection (0.1961) slightly underperforms the Gaussian baseline (0.2030). This result indicates that striped structured sparsity is not universally beneficial and is sensitive to the dataset's topology and label distribution. Flickr is a large social graph (89K nodes) with multi-label classification; the relatively low absolute Macro-F1 values reflect the difficulty of the task.

TABLE IV
NODE CLASSIFICATION — FLICKR (MACRO-F1)

Method	Macro-F1
Gaussian Baseline	0.2030
Striped Structured	0.1961
Hybrid	*N/A (no node features)*

B. Link Prediction

Table V reports link prediction performance (AUC and MRR@10) on Cora, Citeseer, and PubMed using the best hyperparameters selected by Optuna on the validation split. Results for the Gaussian baseline and the Hybrid projection are reported; the striped-only variant was not evaluated separately in the link prediction experiments.

TABLE V
LINK PREDICTION PERFORMANCE (AUC / MRR@10)

Variant	Cora	Citeseer	PubMed
Gaussian	84.50% / 43.29%	77.19% / 39.08%	78.22% / 43.49%
Hybrid	**87.90% / 44.08%**	**81.08% / 42.17%**	**85.09%** / 15.24%

Table VI reports mean embedding CPU time per variant. Values are wall-clock process times measured by `time.process_time()`.

TABLE VI
EMBEDDING CPU TIME FOR LINK PREDICTION (SECONDS)

Variant	Cora	Citeseer	PubMed
Gaussian	0.090	0.057	1.434
Hybrid	0.122	0.090	0.970

C. Discussion

1) Why Hybrid Projections Help on Feature-Rich Graphs: The most striking finding of this study is the 16.8 percentage-point improvement in test accuracy achieved by the hybrid projection on ogbn-arxiv over both structure-only baselines (Table III). This result deserves careful interpretation. The Gaussian and striped projections assign random coordinates to each node before graph propagation; because these coordinates carry no semantic content, the only signal available to downstream classification comes from structural proximity after aggregation. Graph topology alone is insufficient to distinguish the 40 subject-area categories in ogbn-arxiv, where many papers from different fields are cited by similar hub papers.

The hybrid projection allocates $d_{\text{feat}} = d/4$ of the embedding budget to a random projection of the 128-dimensional node features, which are paper title and abstract embeddings generated by skip-gram word vectors. Even after a single feature projection step, these dimensions encode semantic similarity: two papers in the same sub-field will have nearby feature projections, and the subsequent graph propagation reinforces this alignment by aggregating projections from structurally similar neighbours. This combination of semantic initialisation and structural smoothing produces a considerably more discriminative embedding than either component alone, a mechanism that echoes the motivation for combining explicit node attributes with neighbourhood information in methods such as GraphSAGE [1] and modern GNN surveys [2].

A similar, if smaller, benefit holds on Cora (Macro-F1: 0.7541 vs. 0.7232 Gaussian), where bag-of-words features provide lexically distinct signals for the seven paper categories. The fact that the gain on Cora is smaller than on ogbn-arxiv is consistent with this interpretation: Cora's graph is small and tightly clustered, so structural propagation already recovers much of the class signal even without feature information.

2) Why Structured Sparsity Is Dataset-Sensitive: The striped projection outperforms the Gaussian baseline on Cora (0.7356 vs. 0.7232) but trails it on Flickr (0.1961 vs. 0.2030). This divergence sheds light on the conditions under which structured sparsity is beneficial.

On Cora, the graph is sparse and exhibits clear community structure aligned with paper categories. In this setting, the group-wise constraint of the striped projection—exactly one non-zero entry per group of g dimensions— acts as a form of regularisation that reduces inter-dimension correlation in the initial representation, potentially improving the geometry of the embedding before propagation. The JL lemma guarantees that such sparse matrices preserve pairwise distances [7], [9], and the structured sparsity may reduce the effective noise level in the initial embedding, giving graph propagation a cleaner signal to amplify.

On Flickr, the graph is much denser (approximately $10\times$ more edges per node than Cora) and the task involves 195-class multi-label classification, where the decision boundaries are spread diffusely across the feature space. In such settings, the group-wise constraint of the striped matrix may introduce correlated dead zones across embedding dimensions—if a

group's non-zero entry happens to be uninformative for a particular node, all $g-1$ remaining dimensions carry zero signal for that node. This pathology is analogous to the degradation documented by Tadíć et al. [8] for degree-skewed graphs: structure imposed on a projection that was not designed for the graph's topology can actively harm embedding quality. Together these observations suggest that the group size g should be calibrated to the graph's density and the task's label complexity.

3) Link Prediction: Hybrid Consistently Outperforms Gaussian: Across all three citation benchmarks (Cora, Citeseer, PubMed), the hybrid projection achieves higher AUC than the Gaussian baseline (Table V), with improvements of 3.40, 3.89, and 6.87 percentage points respectively. The node features in these datasets (bag-of-words for Cora and Citeseer; TF-IDF for PubMed) provide semantically meaningful similarity information: two papers that share many keywords are likely to be related and, therefore, potentially linked. By incorporating feature similarity into the initial projection, the hybrid embedding makes the dot-product link score a better approximation of true citation probability than a structure-only score.

The one anomalous result is PubMed's MRR@10 for the hybrid (15.24% vs. 43.49% for Gaussian). AUC and MRR@10 measure complementary aspects of ranking quality: AUC measures global discrimination while MRR@10 rewards having at least one true positive in the top-10 ranked candidates per query. The large MRR@10 gap suggests that, despite achieving high overall AUC, the hybrid embedding fails to place true positive links in the very top positions for many PubMed query nodes. One plausible explanation is that PubMed's feature space (TF-IDF over biomedical vocabulary) is high-dimensional and sparse, so the feature projection—which uses a Gaussian matrix scaled by $1/F$—may require more than $d/4$ dimensions to faithfully represent biomedical relatedness, leading to imprecise top-k ranking even when global AUC is better. This interpretation motivates future experiments that vary the structure-to-feature split ratio beyond the fixed 3:1 used here.

4) Computational Efficiency: A key design goal of projection-based embeddings is to eliminate the backpropagation cost of end-to-end GNNs. All three methods in this study satisfy this goal: no gradients are computed, and embedding generation requires a single matrix multiplication per propagation step. The embedding CPU times reported in Table VI confirm that all variants finish in well under two seconds even on PubMed (19.7K nodes).

A notable result is that the hybrid projection is *faster* than the Gaussian baseline on PubMed (0.970 s vs. 1.434 s). This counterintuitive outcome arises because the hybrid replaces $d/4$ of the full-graph structural projection dimensions with a feature projection step; the feature projection is a dense matrix multiply of size $N \times F$ times $F \times d/4$, but $F = 500$ is small relative to $N = 19{,}717$, making this operation cheaper than the corresponding graph propagation over 500 dimensions. On smaller graphs (Cora: 2,708 nodes, Citeseer: 3,327 nodes), this substitution is less impactful and the hybrid incurs a modest overhead of at most 35 ms due to the additional feature projection multiply. These results are consistent with the memory-efficiency findings reported for RAP-GNN [3] and RpHGNN [17]: carefully designed projection operators can simultaneously improve accuracy *and* reduce runtime relative to naive dense baselines.

A full comparison with end-to-end GNN runtimes (GCN, GraphSAGE, GAT) is beyond the scope of this study—such models require multiple training epochs and GPU infrastructure—but the single-forward-pass nature of the present approach suggests it could serve as a fast initialisation stage before fine-tuning, following the spirit of RP-forest GCN initialisation [18].

5) Limitations and Threats to Validity: **Absence of strong GNN baselines.** This study does not compare against end-to-end trained GNNs such as GCN, GraphSAGE [1], or GAT. The primary claim is a controlled comparison *across projection types*, not a claim that projection-based methods are superior to learned methods in absolute terms. Including GNN baselines is important future work for situating these results in the broader landscape.

Fixed structure-to-feature split. The 3:1 structure-to-feature dimension ratio in the hybrid projection was chosen empirically without systematic ablation. Different ratios may be optimal for different datasets or tasks, and the anomalous PubMed MRR@10 result specifically suggests that a larger feature share may be beneficial for high-dimensional sparse feature spaces.

Single projection trial. Unlike the link prediction pipeline (which averages over five random seeds), node classification results for a given hyperparameter configuration reflect a single random seed for the projection matrix. Variability across projection seeds could affect Macro-F1 by a small amount, particularly for Flickr where the margin between Gaussian and striped is only 0.0069.

Homogeneous graph assumption. All datasets used are homogeneous (single node and edge type). The hybrid projection has not been evaluated on heterogeneous graphs, where the distinction between node types would require type-conditioned projection strategies as used in RpHGNN [17].

Transductive setting. All node classification experiments operate in the transductive setting (all nodes visible at embedding time). Inductive extensions, where test nodes are unseen during embedding, require additional study.

VI. Conclusion

This paper investigated the effect of projection matrix design on the quality of graph embeddings within a FastRP-style framework. Three strategies were compared—a dense Gaussian baseline, a striped structured sparse projection, and a hybrid structure-feature projection—across node classification and link prediction tasks on seven benchmark datasets.

The results establish that the hybrid structure-feature projection consistently achieves the best predictive performance on datasets with informative node attributes, with particularly large gains on the large-scale ogbn-arxiv benchmark (16.8

percentage point improvement in test accuracy). The striped structured projection demonstrates competitive performance on citation graphs, supporting the viability of sparse structured projections as a scalable alternative to dense Gaussian projections. However, its sensitivity on the Flickr social graph underscores the importance of dataset-aware hyperparameter tuning, particularly the group size g.

Future work should explore: (i) a systematic ablation of the structure-to-feature dimension ratio in the hybrid projection; (ii) integration of these projection strategies into full GNN architectures as a preprocessing or regularization component; (iii) extension to heterogeneous and temporal graphs; and (iv) a detailed runtime and memory efficiency characterization across varying graph sizes.

Author Contributions

H.A.D.T.N. Haththella contributed to the design of the projection strategies, implementation of the striped structured projection, and the node classification experiments. L.R.S.R. Premasiri contributed to the implementation of the hybrid projection, the link prediction pipeline, and hyperparameter tuning with Optuna. A.N.T. Somathilaka contributed to dataset preparation, data loading infrastructure, and experimental analysis. All authors participated in drafting and revising the manuscript. Dr. Sampath Deegalla supervised the project.

References

[1] B. Khemani, S. Patil, K. Kotecha, and S. Tanwar, "A review of graph neural networks: concepts, architectures, techniques, challenges, datasets, applications, and future directions," *Journal of Big Data*, vol. 11, no. 1, p. 18, 2024.

[2] G. Gkarmpounis, C. Vranis, N. Vretos, and P. Daras, "Survey on graph neural networks," *IEEE Access*, vol. 12, pp. 128 816–128 832, 2024.

[3] T. Bui, C.-B. Schönlieb, B. Ribeiro, B. Bevilacqua, and M. Eliasof, "On the effectiveness of random weights in graph neural networks," *arXiv preprint arXiv:2502.00190*, 2025.

[4] H. Chen, S. F. Sultan, Y. Tian, M. Chen, and S. Skiena, "Fast and accurate network embeddings via very sparse random projection," in *arXiv preprint arXiv:1908.11512*, 2019.

[5] Z. Zhang, P. Cui, H. Li, X. Wang, and W. Zhu, "Billion-scale network embedding with iterative random projection," *arXiv preprint arXiv:1805.02396*, 2018.

[6] Y. Xie, Y. Dong, J. Qiu, W. Yu, X. Feng, and J. Tang, "Sketchne: Embedding billion-scale networks accurately in one hour," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 10, pp. 10 666–10 680, 2023.

[7] W. B. Johnson and J. Lindenstrauss, "Extensions of lipschitz mappings into a hilbert space," *Contemporary Mathematics*, vol. 26, pp. 189–206, 1984.

[8] T. Tadić, C. Becker, and J. Neville, "Node similarities under random projections: Limits and pathological cases," *arXiv preprint arXiv:2404.10148*, 2024.

[9] D. Achlioptas, "Database-friendly random projections: Johnson-lindenstrauss with binary coins," *Journal of Computer and System Sciences*, vol. 66, no. 4, pp. 671–687, 2003.

[10] B. Perozzi, R. Al-Rfou, and S. Skiena, "Deepwalk: Online learning of social representations," in *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2014, pp. 701–710.

[11] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 855–864.

[12] J. Tang, M. Qu, M. Wang, M. Zhang, J. Yan, and Q. Mei, "Line: Large-scale information network embedding," in *Proceedings of the 24th International Conference on World Wide Web*, 2015, pp. 1067–1077.

[13] J. Qiu, Y. Dong, H. Ma, J. Li, C. Wang, K. Wang, and J. Tang, "Netsmf: Large-scale network embedding as sparse matrix factorization," in *The World Wide Web Conference*, 2019, pp. 1509–1520.

[14] M. Ou, P. Cui, J. Pei, Z. Zhang, and W. Zhu, "Asymmetric transitivity preserving graph embedding," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1105–1114.

[15] Y. Yahaya, J. O. Ajadi, R. A. Sanusi, Z. Sawlan, and N. A. Adegoke, "Hybrid random projection technique for enhanced representation in high-dimensional data," *Expert Systems with Applications*, vol. 262, p. 125569, 2025.

[16] Q. Wang, J. Wan, F. Nie, B. Liu, C. Yan, and X. Li, "Hierarchical feature selection for random projection," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 30, no. 5, pp. 1581–1586, 2019.

[17] J. Hu, B. Hooi, and B. He, "Efficient heterogeneous graph learning via random projection," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 12, pp. 8093–8107, 2024.

[18] M. Alshammari, J. Stavrakakis, A. F. Ahmed, and M. Takatsuka, "Random projection forest initialization for graph convolutional networks," *MethodsX*, vol. 11, p. 102315, 2023.

[19] P. Li, T. J. Hastie, and K. W. Church, "Very sparse random projections," in *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2006, pp. 287–296.

[20] J. W. Tukey, "On the comparative anatomy of transformations," *The Annals of Mathematical Statistics*, vol. 28, no. 3, pp. 602–632, 1957.

[21] S. Dasgupta and A. Gupta, "An elementary proof of the johnson-lindenstrauss lemma," International Computer Science Institute, Tech. Rep. TR-99-006, 1999.

[22] W. Hu, M. Fey, M. Zitnik, Y. Dong, H. Ren, B. Liu, M. Catanza, and J. Leskovec, "Open graph benchmark: Datasets for machine learning on graphs," in *Advances in Neural Information Processing Systems*, 2020.

[23] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.